

Applied A.I. Solutions

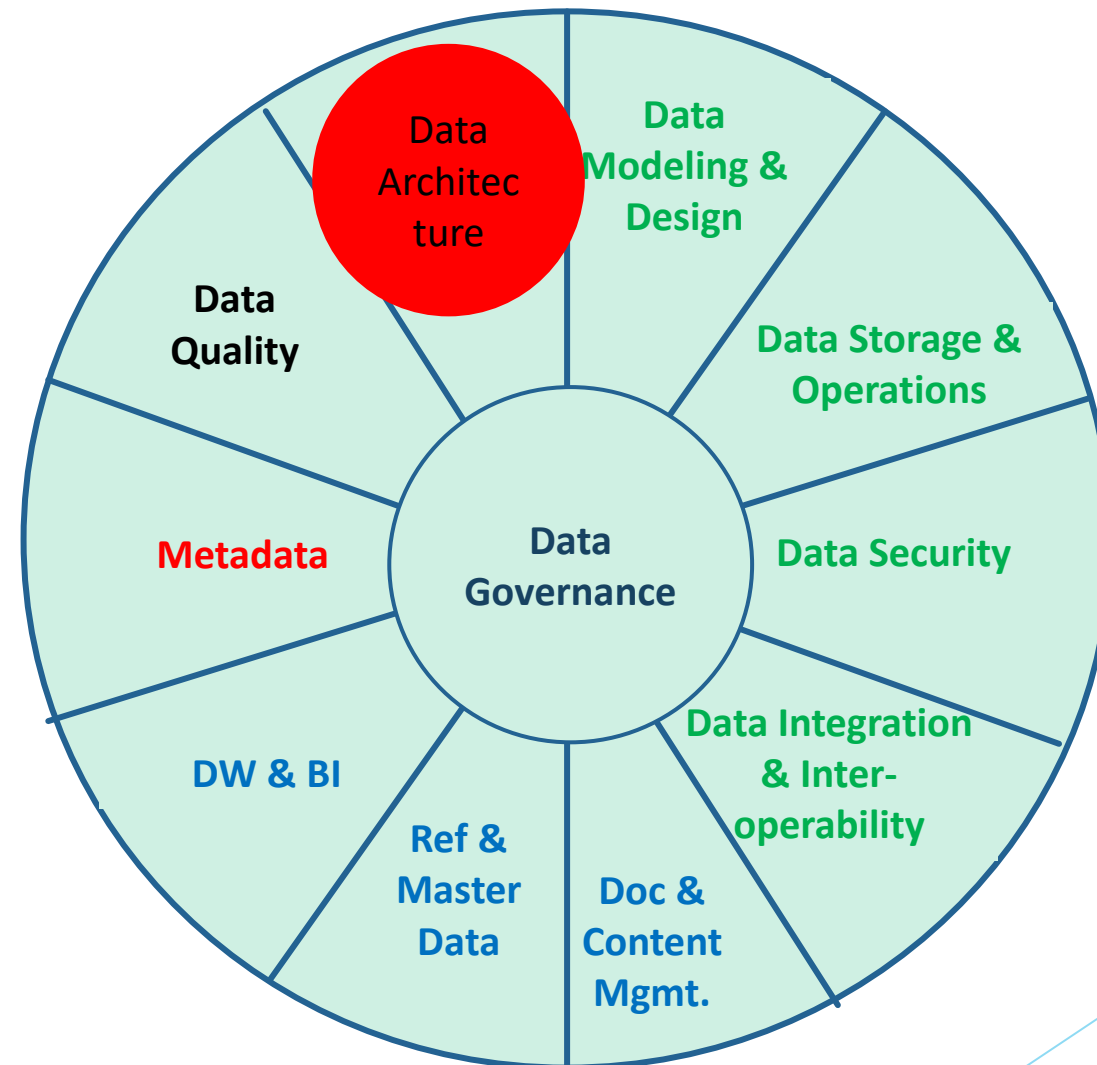
Foundations of Data Management

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DATA ARCHITECTURE

The DAMA Wheel



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Introduction

- Architecture refers to the art and science of **building things** and to its result
- **Enterprise Architecture** encompasses domain architecture, including **business, data, application and technology**

Introduction - Cont'd

- **Data Architecture is fundamental to Data Management**
- The **master design documents** represents data at different levels of abstraction
- Provides a **formal data model**, with data names, comprehensive data, Metadata definitions, conceptual and logical entities and relationships, and business rules
- Provides **standards** that governs how data is collected, stored, arranged, used and removed

Definition

- To design and maintain the **master blueprints** to meet identified **enterprise** data needs
- Using the master blueprints, to guide **data integration, control** data assets, and **align** data investments with business strategy

Business Drivers

- **Readiness** to take advantage of new business opportunities and new technologies
- **Fulfillment** of business process requirements regarding data
- Manage data and information delivery in the **enterprise**

Goals

- **Design structures and plans** to meet current and future data requirements
- Strategically prepare the organization to quickly take advantage of new business opportunities



Business Drivers

Inputs

- **Data Strategies**
- Enterprise Architecture
- Business Architecture
- IT Standards, Goals



Activities

Establish **Enterprise Data Architecture**

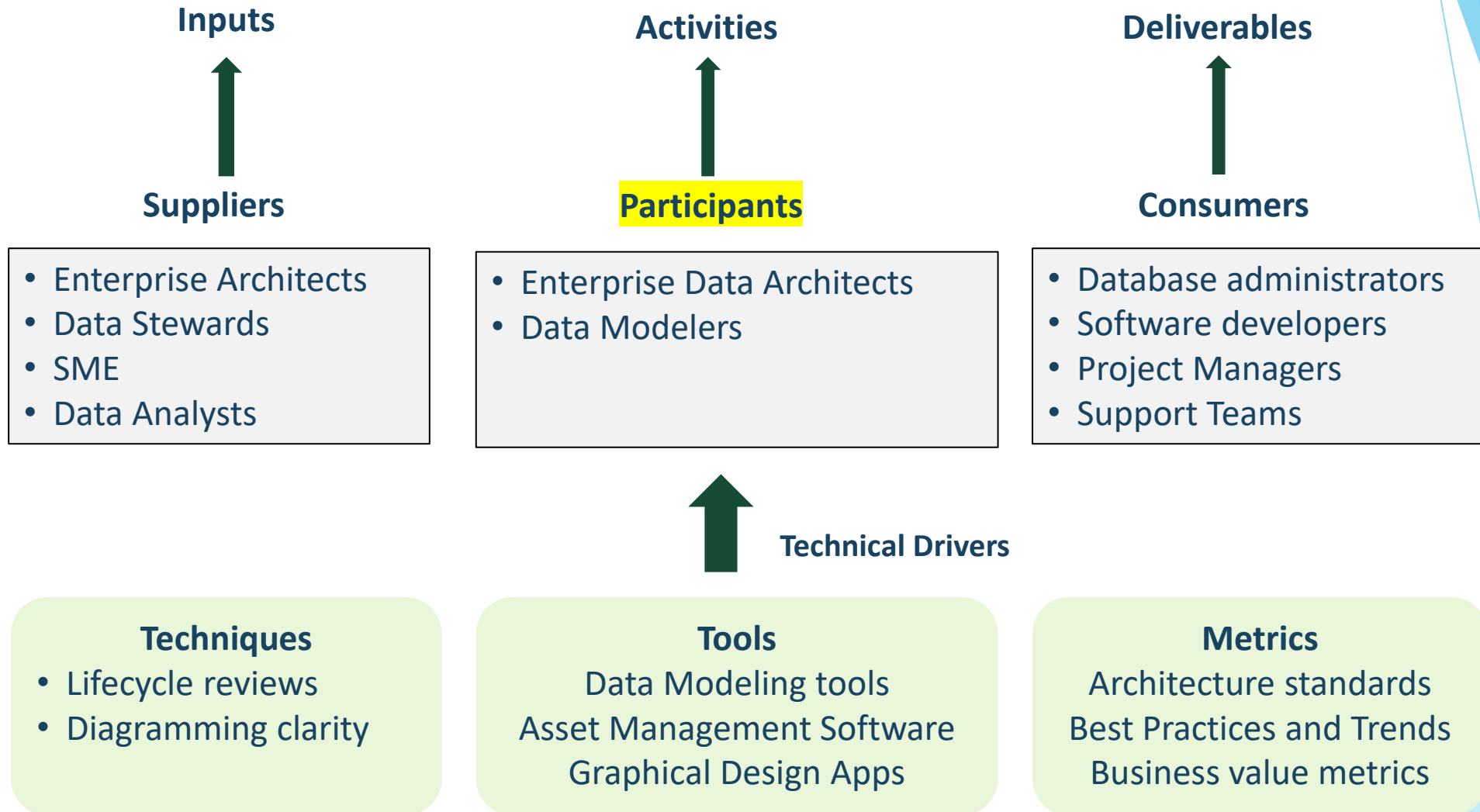
- Evaluate existing data architecture
- Develop a Roadmap
- Manage enterprise requirements

Integrate with **Enterprise Architecture**



Deliverables

- **Data Architecture**
- Data Flows
- Data Value Chains
- Enterprise Data Model
- Implementation Roadmap



Architecture Domains

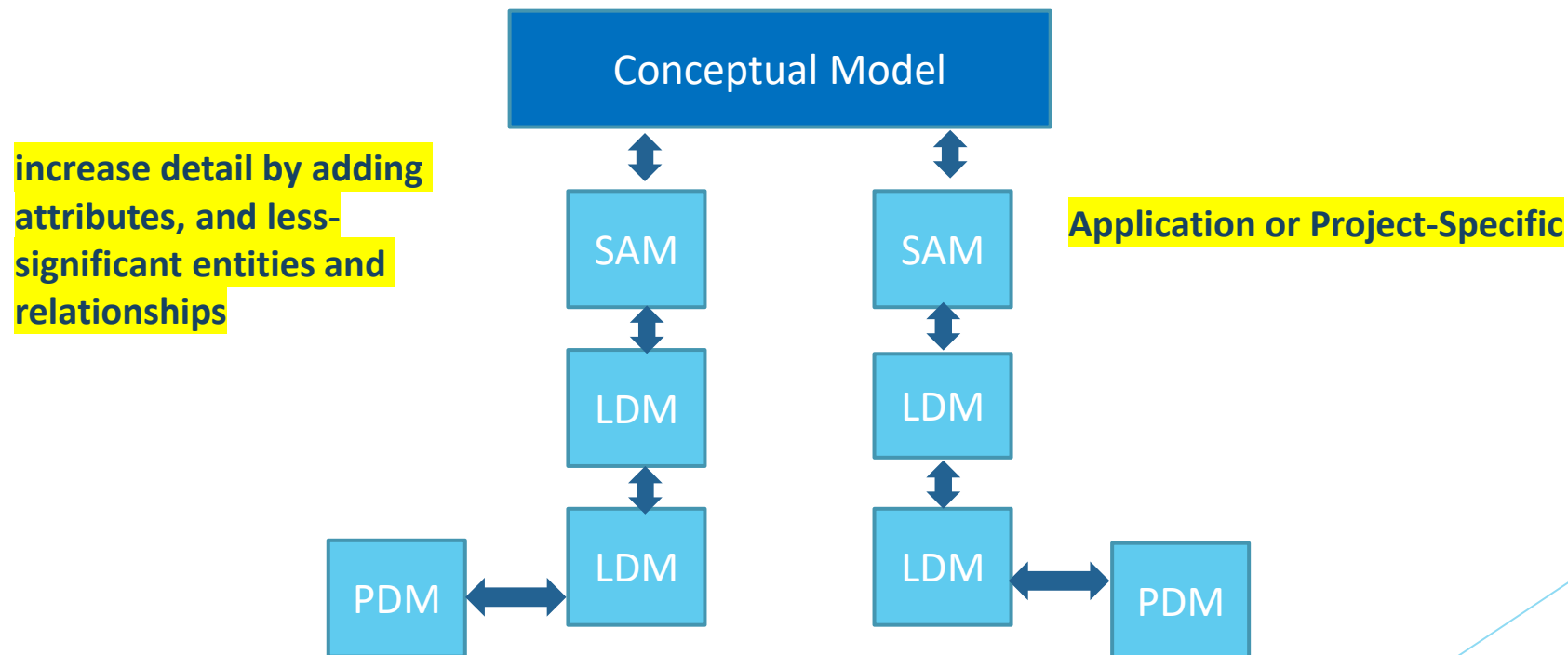
Domain	Business Architecture	Data Architecture	Application Architecture	Technical Architecture
Purpose	To create value	To describe how data should be organized, managed	To describe app structure and functionality	To describe the physical tech to deliver value
Elements	Business models, processes, strategies, capabilities, vocabulary	Data models, data definitions, data mapping, specs, dataflows, APIs	Business systems, SW packages, databases	Technical platforms, networks, security, integration tools
Dependencies	Establishes req for other domains	Manages data created and required by EBA	Acts on specified data according to business req	Hosts, executes the E App A
Roles	Business Architects, Business Analysts, Data Stewards	Data Architects, Data Modelers, Data Stewards	Application Architects	Infrastructure architects

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Enterprise Data Model (EDM)

- It is a **conceptual or logical data model providing a common consistent view of data across the enterprise**
- It includes key enterprise data entities, their relationship, mapping, specs, business rules and critical attributes
- It is the **foundation for all data and data-related projects**
- It can be **built incrementally and iteratively**, using layers

- A **conceptual** view over the enterprise's subject areas
- Views of entities and relationships for each subject area
- **Detailed, partially attributed logical views** of the same subject area
- Logical and physical models specific to an application or project



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Models

EDA deals with two complexities: a) **Quality**, and b) **Innovation**

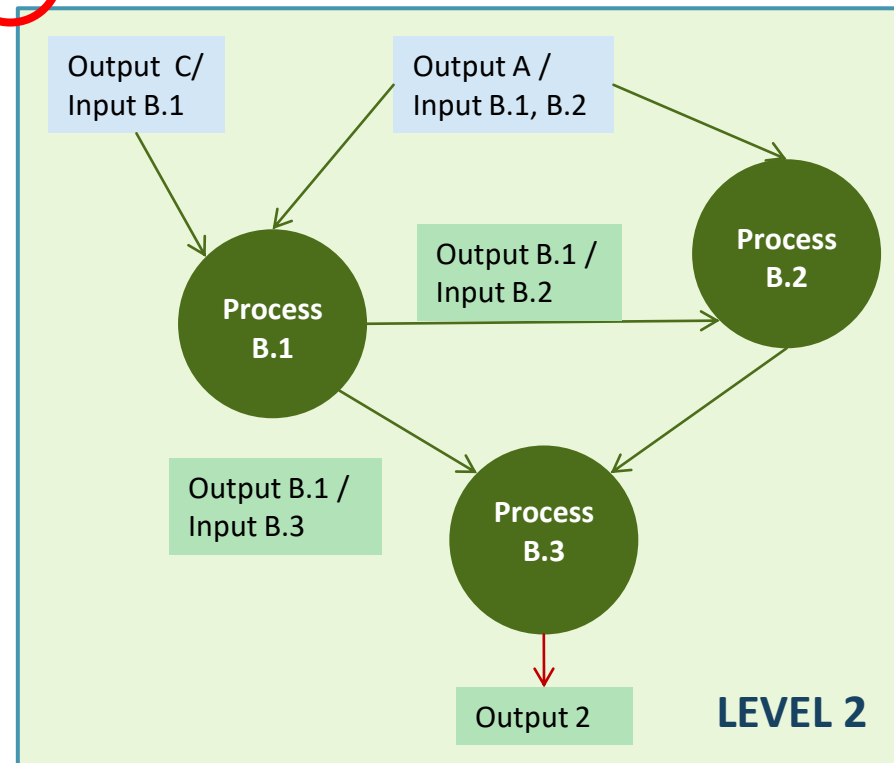
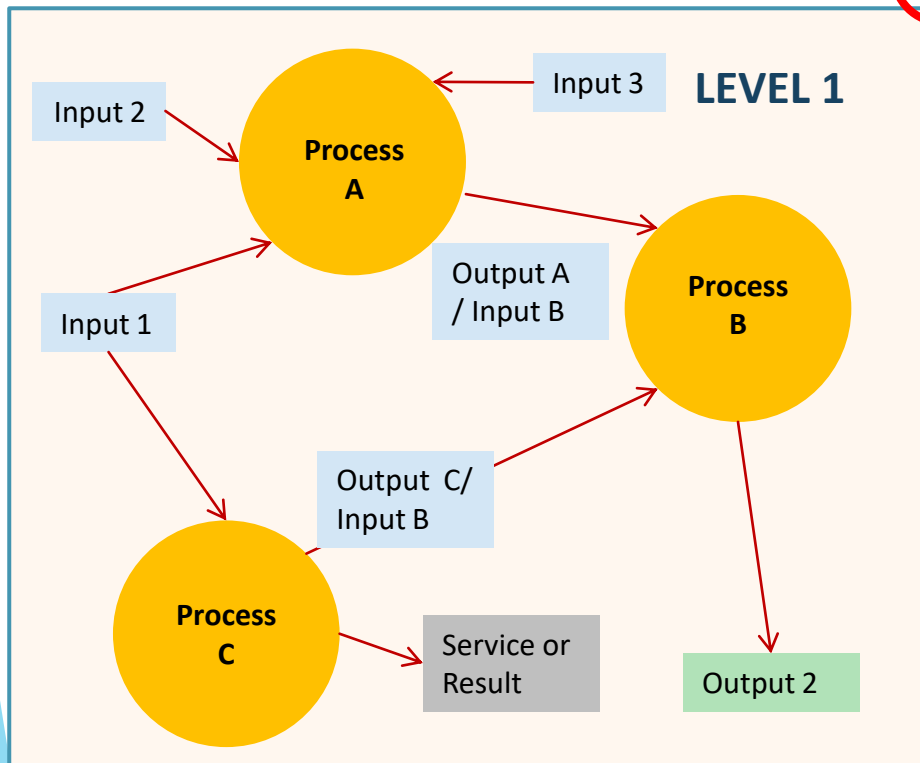
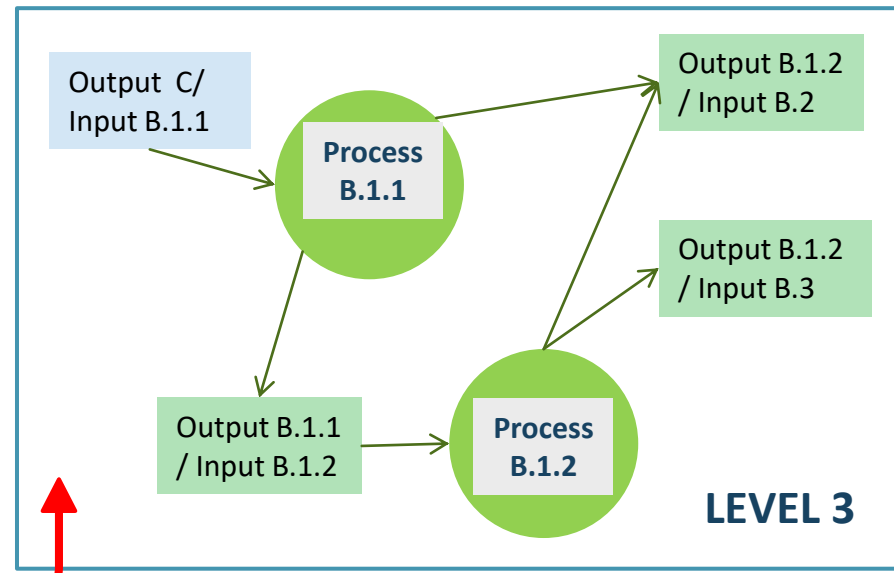
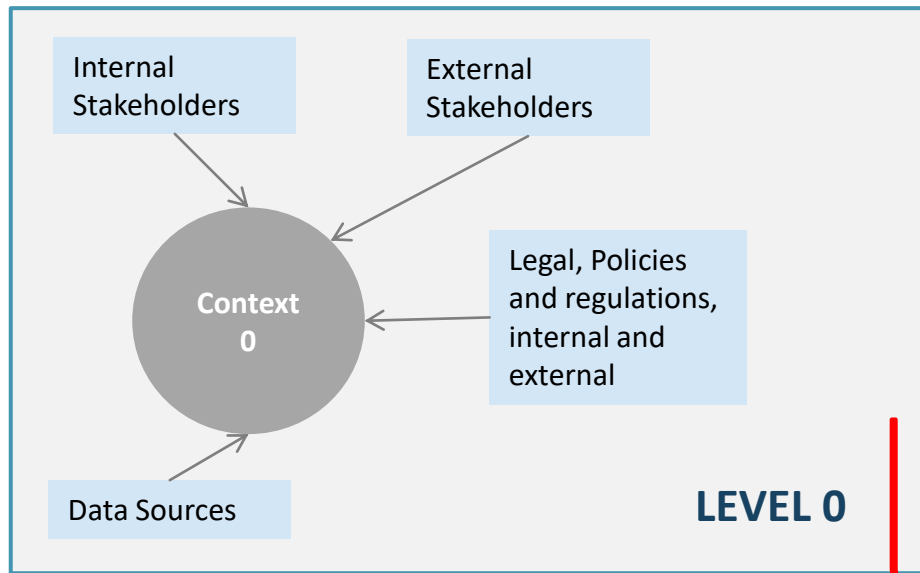
EDA practice includes:

- Levels
↓
- Strategy (framework, roadmap)
 - Acceptance and culture (behaviour)
 - Organization (accountabilities)
 - Working methods (rules)
 - Results (data architecture)

Business Processes

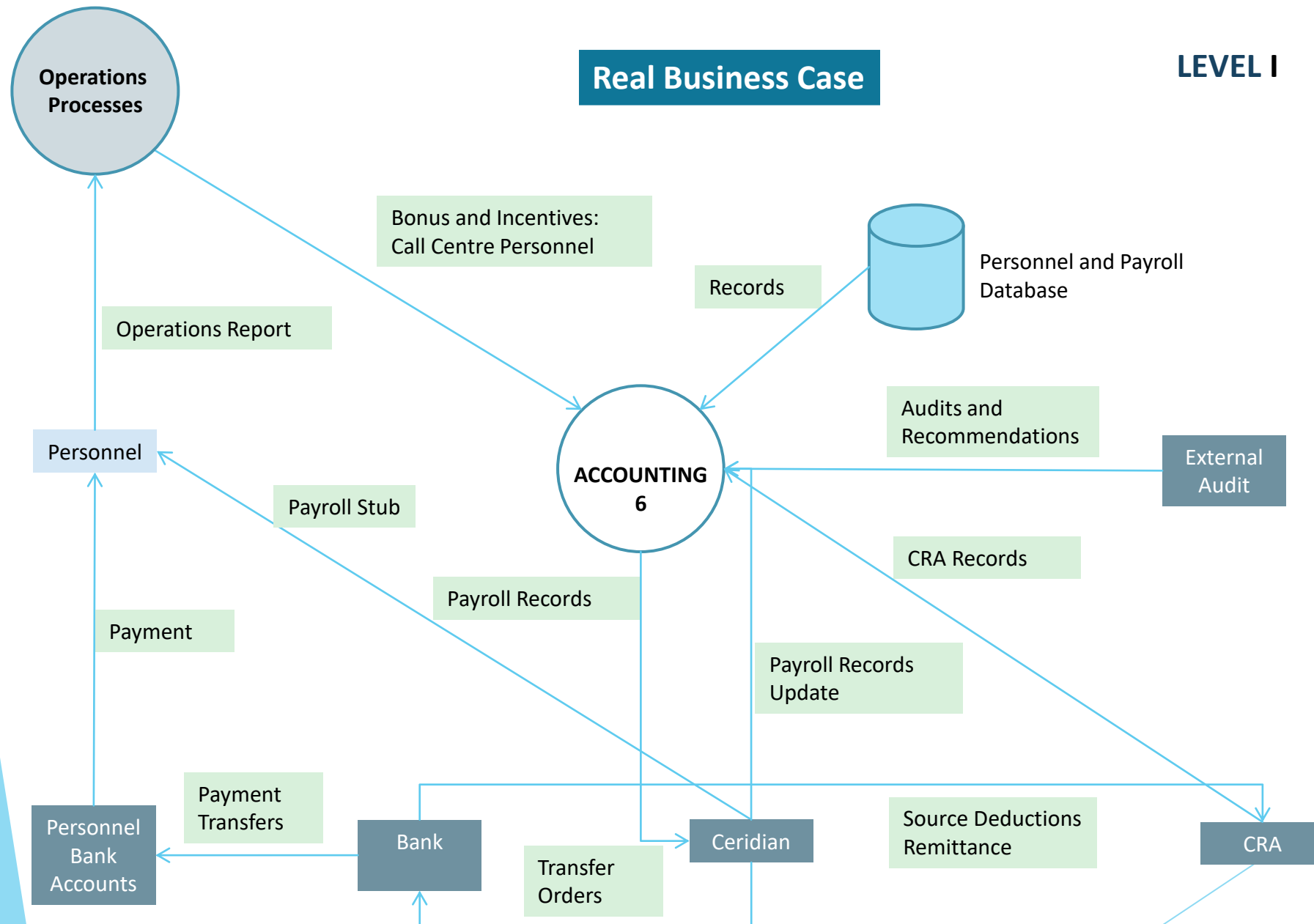
- A process is a set of interrelated actions and activities performed to achieve a pre-specified product, result, or service
- Each process is characterized by its input, the tools and techniques that can be applied, and the resulting outputs
- **Each process can be broken down into other processes**, and so on, until we reach a level in which simple activities are defined using measurable inputs and **measurable outputs**

Automation
AI/ML



Real Business Case

LEVEL I



Data / Process Flow Diagram (design)

- Defines the requirements and master blueprint for data storage and processing across databases, applications, platforms, and networks
- Data flows **map the movement of data** to business processes, locations, business roles (CRUD(E)) and technical components

A blue callout bubble with a tail pointing towards the underlined text 'movement of data' in the list item above. The bubble contains the text 'DATA + METADATA' in bold black letters on a yellow background.

DATA + METADATA

Traceability

All functional levels are part of the Enterprise Data Model, and the linkages create paths to trace data from top to bottom and between models.

- Vertical: models in each level map to models in other levels
- Horizontal: data and data relationships may appear in multiple models in the same level

Orchestration

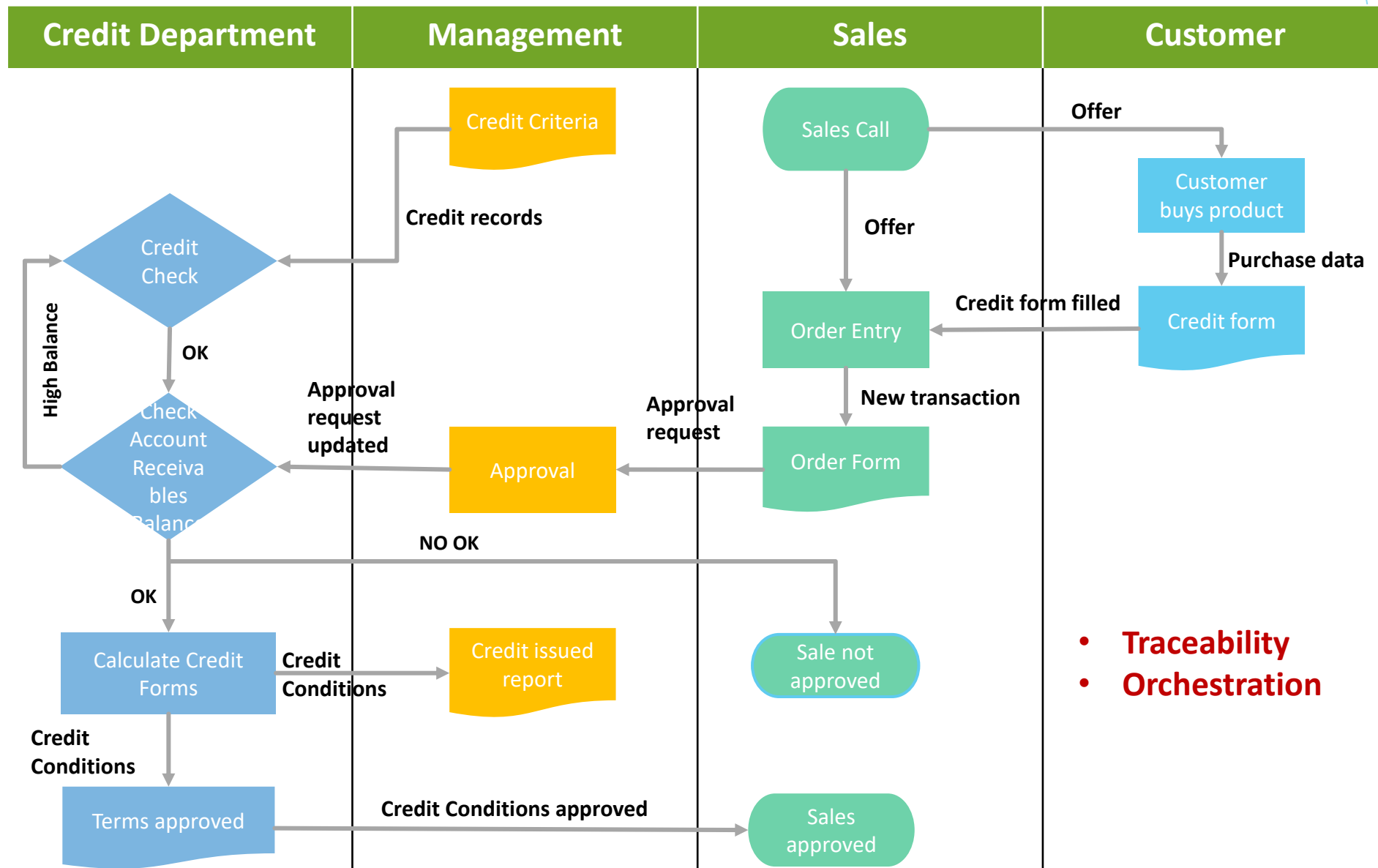
Management approach that connects workflows, technology, and people into end-to-end automated processes.

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Traceability and Orchestration are crucial for AI systems for several reasons:

- **Integration of Diverse Tools:** AI orchestration ensures that various AI tools and models can work together seamlessly, allowing different AI components to communicate, share data, and function as a cohesive system.
- **Efficiency and Effectiveness:** traceability and orchestration streamlines workflows, reduces redundancy, and optimizes resource usage.
- **Scalability:** easier to scale AI systems. As new AI models are added, orchestration ensures they integrate smoothly with existing components, allowing the system to grow and adapt.
- **Improved Performance:** Coordinating the various stages of AI application development, from data processing to model deployment, enhances the overall performance and responsiveness of AI systems.
- **Enhanced Monitoring and Management:** traceability and orchestration provides a framework for monitoring and managing AI processes, ensuring that all components are functioning efficiently.

Data + Process Flow Diagram (design)



- **Traceability**
- **Orchestration**

